# DOCUMENTATION OF GRIFFINDO TOYS PAYROLL SYSTEM

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## The GUI System + Coding

### The Login Interface

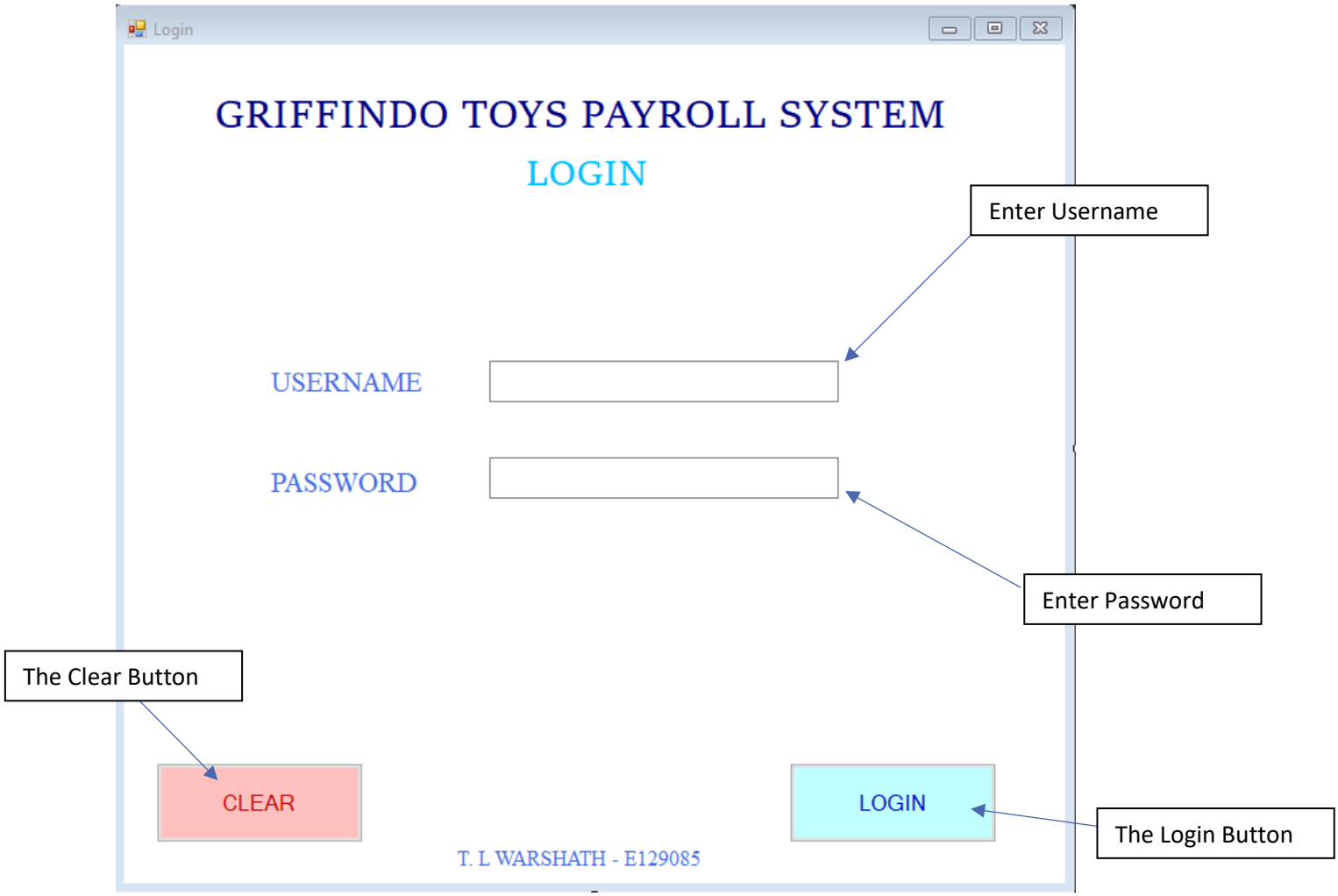



*Figure 1 - The Login page*

This is the first interface where the user will encounter when executing the program. The user needs to gain access to the payroll system by entering a valid username and password. The admins can edit the passwords for this windows form via the SQL database.

## Login Button Code

```
1 reference
private void button2_Click(object sender, EventArgs e)
{

    SqlConnection con = new SqlConnection(@"Data Source=DESKTOP-EQH9955\SQLEXPRESS;Initial Catalog=griffindo_toys;Integrated Security=True");
    string query = "select * from logint where Username = '" + unamet.Text + "' and Password = '" + pwordt.Text + "'";
    SqlDataAdapter sda = new SqlDataAdapter(query, con);
    DataTable dtbl = new DataTable();
    sda.Fill(dtbl);
    if (dtbl.Rows.Count == 1)
    {
        Form2 goo = new Form2();
        this.Hide();
        goo.Show();

    }
    else
    {
        MessageBox.Show("Invalid username and Password");
    }

}
```

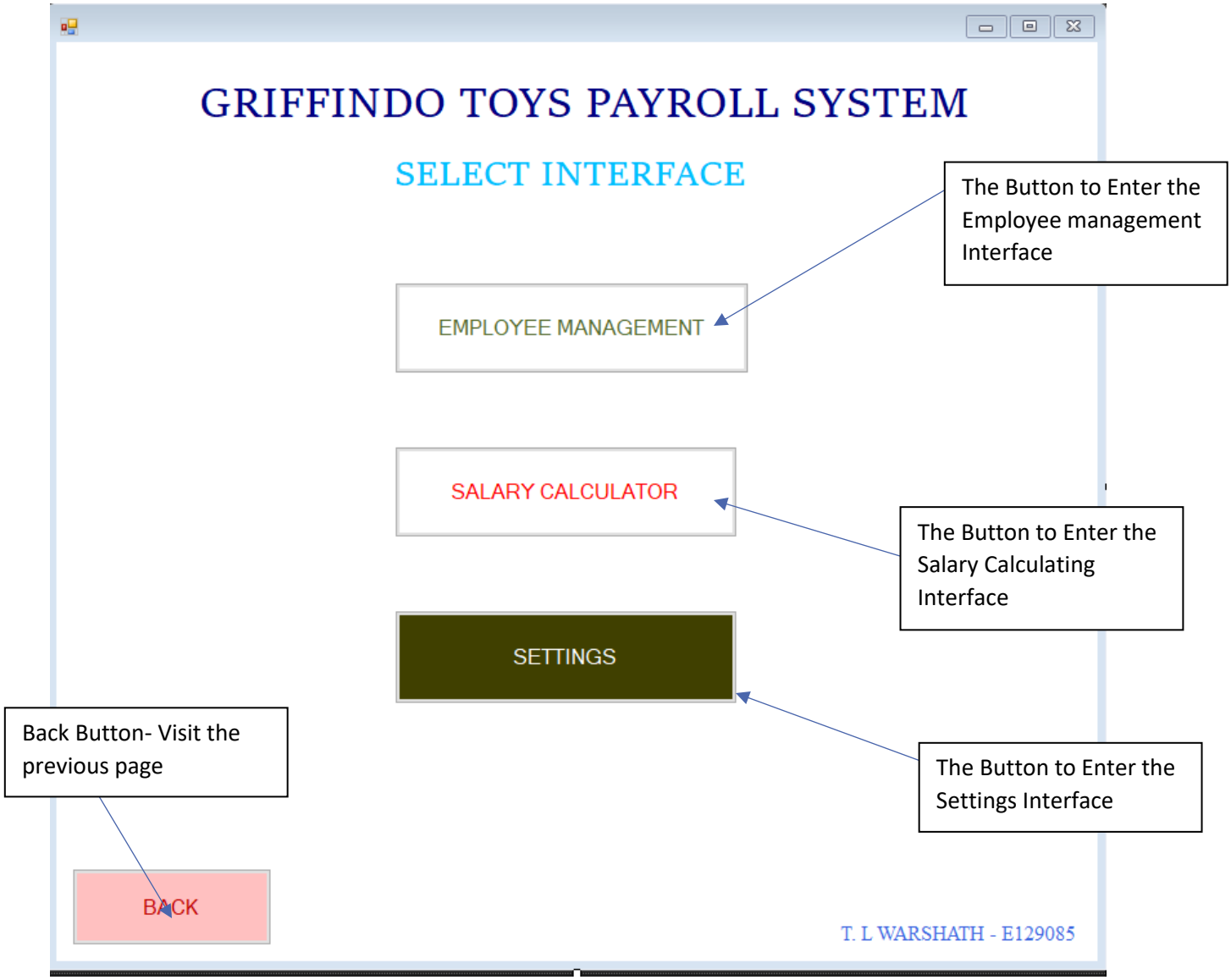
*Figure 2 - login Button Code*

## Clear Button Code

```
1 reference
private void button1_Click(object sender, EventArgs e)
{
    unamet.Clear();
    pwordt.Clear();
}
```

*Figure 3 - Clear Button Code*

## The Selection Interface




In this interface the user can select the interface the user desires to manage. The user can either select to manage employees, calculate the salary of employees or to edit the settings of the system.

## Employee Management Button Code

```
1 reference
private void button2_Click(object sender, EventArgs e)
{
    Form3 goo = new Form3();
    this.Hide();
    goo.Show();
}
```

*Figure 4 - Employee Management Button Code*

## Salary Calculation Button Code

```
1 reference
private void button1_Click(object sender, EventArgs e)
{
    Form4 goo = new Form4();
    this.Hide();
    goo.Show();

}
```

*Figure 5 - Salary Calculation Button Code*

## Settings Button Code

```
1 reference
private void button3_Click(object sender, EventArgs e)
{
    Form5 goo = new Form5();
    this.Hide();
    goo.Show();
}
```

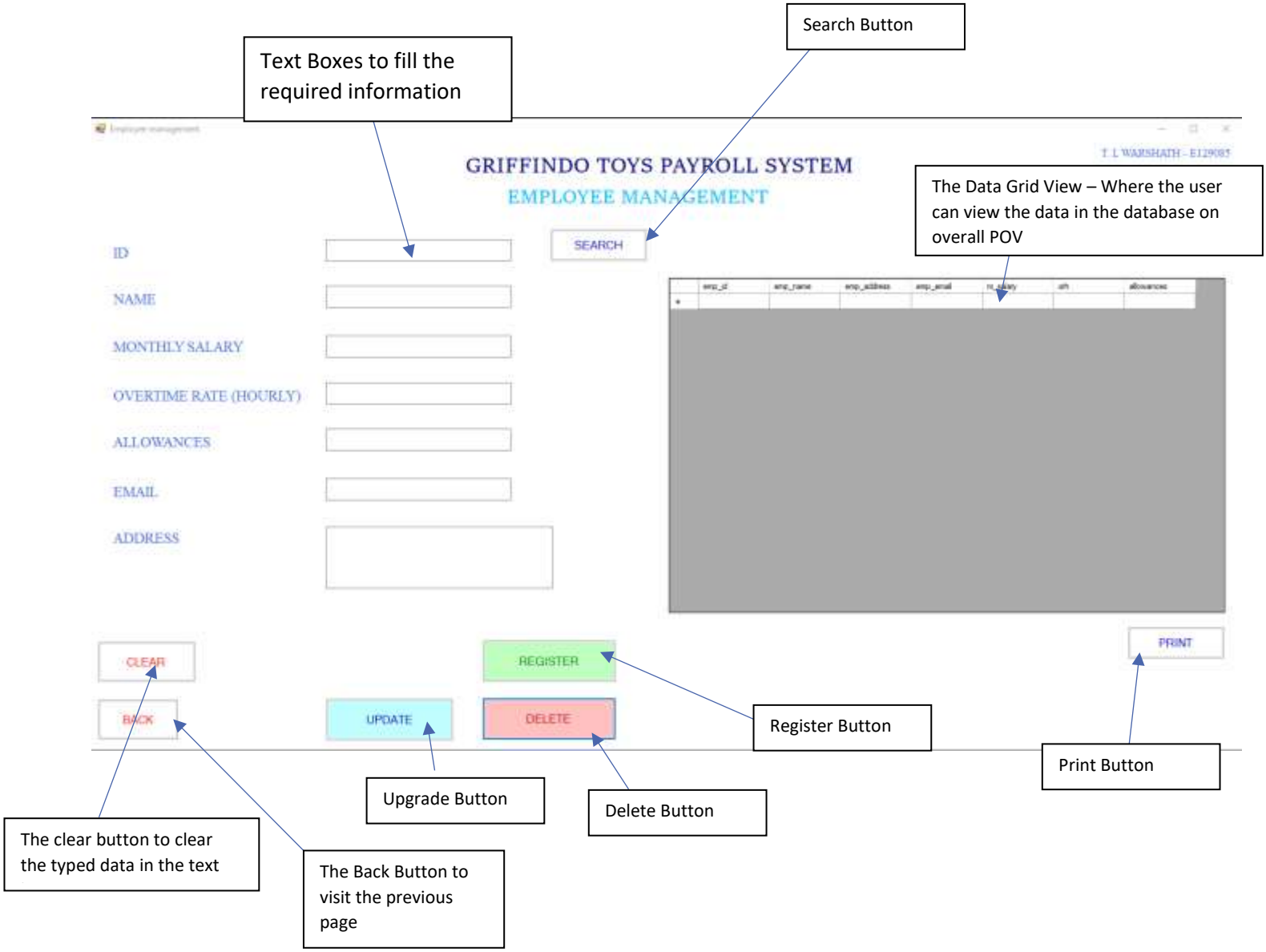
*Figure 6 - Settings Button Code*

## Back Button Code

```
private void button4_Click(object sender, EventArgs e)
{
    Form1 goo = new Form1();
    this.Hide();
    goo.Show();
}
```

*Figure 7- Back Button Code*

## The Employee Management Interface




In this interface the user can edit the employee data. The user can register, Update and delete employee data by filling the above visible text boxes. The user can extract individual employee data by entering the desired employee ID and extract the data to the textboxes. The user also has the ability to view the entire employee table on the right and print the data as required.

## Register Button Code

```
private void button4_Click(object sender, EventArgs e)
        {
            Form2 goo = new Form2();
            this.Hide();
            goo.Show();
        }

        private void button2_Click(object sender, EventArgs e)
        {
            if (empid.Text == "" || name.Text == "" || ms.Text == "" || or.Text == "" || al.Text
== "" || mail.Text == "" || add.Text == "")
            {
                MessageBox.Show("Fill all the boxes and try again");

            }
            else
            {

                try
                {


                    con.Open();
                    string query = "insert into Employee values('" + empid.Text + "','" +
name.Text + "','" + add.Text + "','" +  mail.Text + "','" + ms.Text + "','" +or.Text+ "','" +
al.Text + "')";
                    SqlCommand cmd = new SqlCommand(query, con);
                    cmd.ExecuteNonQuery();
                    MessageBox.Show("Your Data Has Beeen Successfully Added");
                    con.Close();
                    filldatagrid();

                }
                catch (Exception Ex)
                {

                    MessageBox.Show(Ex.Message);

                }
            }
```

*Figure 8 - register Button Code (Employee)*

## Update Button Code

```
private void button1_Click(object sender, EventArgs e)
        {


            try
            {
                con.Open();
                string query = "update Employee set emp_name='" + name.Text + "',m_salary= '" + ms.Text+
"',emp_email='" + mail.Text + "',emp_address='" +add.Text + "',orh='" +or.Text + "',allowances='" +
al.Text+ "' where emp_id='" + empid.Text+ "';";
                SqlCommand cmd = new SqlCommand(query, con);
                cmd.ExecuteNonQuery();
                MessageBox.Show("You Have Successfully Updated Employee Data");
                con.Close();
                filldatagrid();
                cdata();

            }
            catch (Exception Ex)
            {
                MessageBox.Show(Ex.Message);
            }
                                                        }
```

*Figure 9- Update Button Code (Employee)*

## Delete Button Code

```
private void button3_Click(object sender, EventArgs e)
   {
       if (empid.Text == "")
       {
           MessageBox.Show("Please Enter The ID of the Employee You prefer to Delete");
       }
       else
       {
           try
           {
               con.Open();
               string query = "delete from Employee where emp_id ='" + empid.Text + "';";
               SqlCommand cmd = new SqlCommand(query, con);
               cmd.ExecuteNonQuery();
               MessageBox.Show("Employee Data Successfully Deleted");
               con.Close();
               filldatagrid();
               cdata();
           }
           catch (Exception Ex)
           {

               MessageBox.Show(Ex.Message);

           }
       }
                                                   }
```

*Figure 10-Delete Button Code (Employee)*

## Search Button Code

```
1 reference
private void button5_Click(object sender, EventArgs e)
{
    con.Open();
    string query = "select * from Employee where emp_id='" + empid.Text + "'";
    SqlCommand cmd = new SqlCommand(query, con);
    DataTable dt = new DataTable();
    SqlDataAdapter sda = new SqlDataAdapter(cmd);
    sda.Fill(dt);
    foreach (DataRow dr in dt.Rows)
    {

        empid.Text = dr["emp_id"].ToString();
        name.Text = dr["emp_name"].ToString();
        ms.Text = dr["emp_address"].ToString();
        mail.Text = dr["emp_email"].ToString();
        add.Text = dr["emp_address"].ToString();
        al.Text = dr["allowances"].ToString();
        or.Text = dr["orh"].ToString();

        empid.Visible = true;
        name.Visible = true;
        ms.Visible = true;
        mail.Visible = true;
        add.Visible = true;
        al.Visible = true;
        or.Visible = true;


    }
    con.Close();
}
```

*Figure 11- Search Button Code*

## Print Button Code

```
    SqlConnection con = new SqlConnection(@"Data Source=DESKTOP-EQH9955\SQLEXPRESS;Initial
Catalog=griffindo_toys;Integrated Security=True");

        private DataTable dataTable;
-------------------------------------------------------------------------------------------


   private void Form3_Load(object sender, EventArgs e)
        {
             filldatagrid();
             LoadData();
        }



-------------------------------------------------------------------------------------------
    private void button6_Click(object sender, EventArgs e)
        {
            try
            {
                SaveFileDialog saveFileDialog = new SaveFileDialog();
                saveFileDialog.Filter = "CSV Files|*.csv";
                if (saveFileDialog.ShowDialog() == DialogResult.OK)
                {
                    string filePath = saveFileDialog.FileName;
                    ExportDataTableToCsv(dataTable, filePath);
                    MessageBox.Show("Data saved to " + filePath, "Success", MessageBoxButtons.OK,
MessageBoxIcon.Information);
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error: " + ex.Message);
            }

        }
-------------------------------------------------------------------------------------------
        private void ExportDataTableToCsv(DataTable dataTable, string filePath)
        {
            using (System.IO.StreamWriter writer = new System.IO.StreamWriter(filePath))
            {

                string columnHeaders = string.Join(",", dataTable.Columns.Cast<DataColumn>().Select(dc =>
dc.ColumnName));
                writer.WriteLine(columnHeaders);
                foreach (DataRow row in dataTable.Rows)
                {
                    string rowData = string.Join(",", row.ItemArray);
                    writer.WriteLine(rowData);
                }
            }
                                                        }
```

```
private void LoadData()
        {
            try
            {
                {
                    con.Open();
                    string query = "SELECT * FROM Employee";
                    SqlCommand cmd = new SqlCommand(query, con);

                    SqlDataAdapter adapter = new SqlDataAdapter(cmd);
                    dataTable = new DataTable();
                    adapter.Fill(dataTable);
                    dataGridView1.DataSource = dataTable;

                    con.Close();
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error: " + ex.Message);
            }
        }
```

*Figure 12 - Print Button Code*

## Data Grid View – Code

```
    }

    1 reference
    private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
    {
        empid.Text = dataGridView1.SelectedRows[0].Cells[0].Value.ToString();
        name.Text = dataGridView1.SelectedRows[0].Cells[1].Value.ToString();
        ms.Text = dataGridView1.SelectedRows[0].Cells[2].Value.ToString();
        or.Text = dataGridView1.SelectedRows[0].Cells[6].Value.ToString();
        al.Text = dataGridView1.SelectedRows[0].Cells[3].Value.ToString();
        mail.Text = dataGridView1.SelectedRows[0].Cells[5].Value.ToString();
        add.Text = dataGridView1.SelectedRows[0].Cells[7].Value.ToString();
    }
```

*Figure 13 - Data Grid View Code*

## Clear Button Code

```
3 references
private void cdata()
{
    //use to clear the text boxes.
        empid.Clear();
        ms.Clear();
        add.Clear();
        or.Clear();
        al.Clear();
        name.Clear();
        mail.Clear();
```

```
    1 reference
    private void button7_Click(object sender, EventArgs e)
    {
        cdata();
    }
}
```

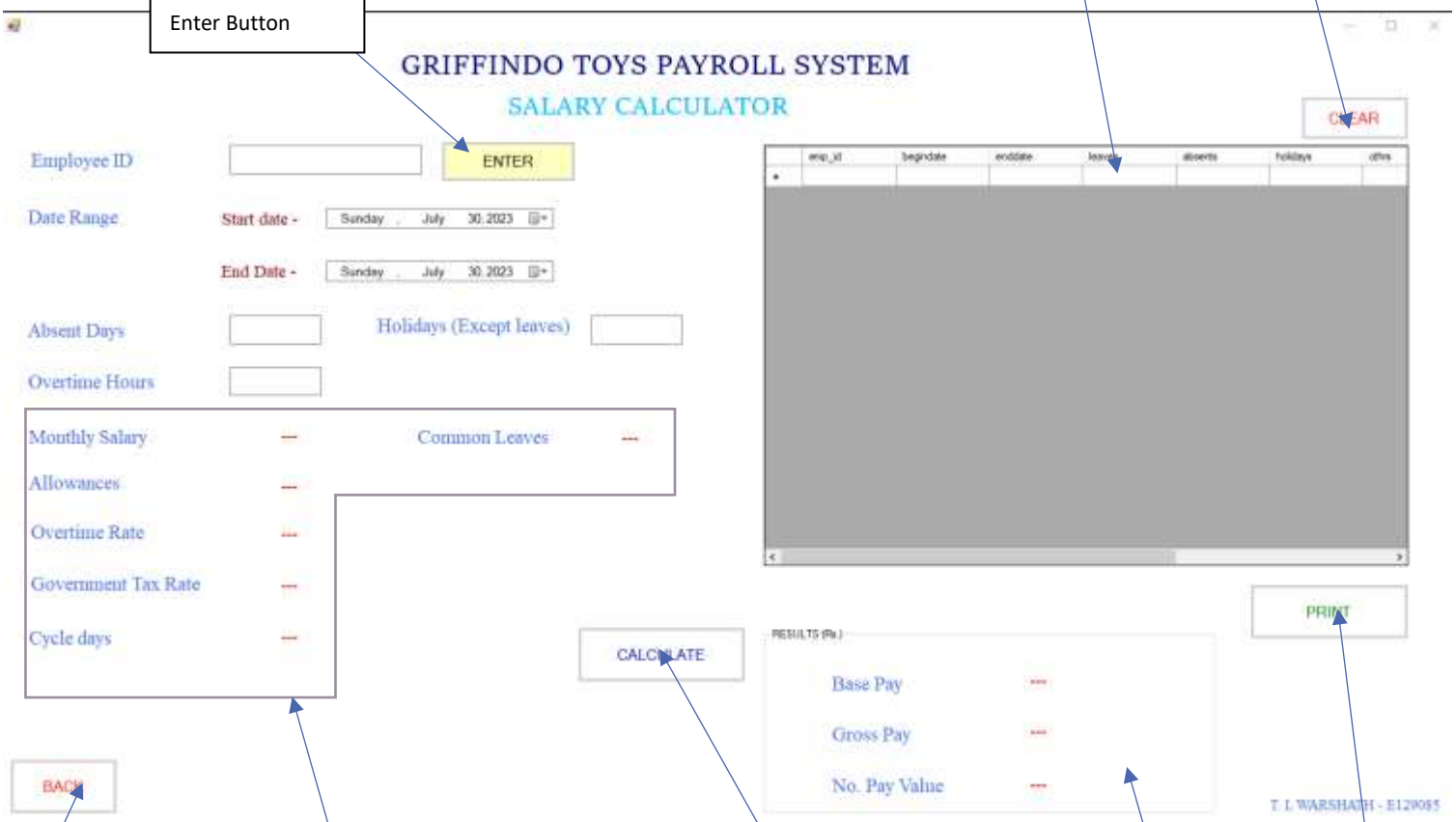
*Figure 14 - The code to clear button*

## Back Button Code

```
1 reference
private void button4_Click(object sender, EventArgs e)
{
    Form2 goo = new Form2();
    this.Hide();
    goo.Show();
}
```

*Figure 15- Back Button Code*

## The Salary Calculator Interface




Clear Button – To clear data in the data grid view

Data Grid View

Enter Button

Back Button

Labels to show the extracted data according to the employee ID when it is entered.

Calculate Button

Results Of calculations

Print Button

Using this interface, the user can calculate the salary and data. The user can input the date range to calculate the salary. Salary cycle begin date and the end date should be given to calculate the monthly salary. All the calculated No-pay-value, Base-pay-value and Gross pay will be recorded in the database (data grid view connected) under each employee for each month. The print button can be used to generate reports such as monthly salary report for an employee, overall salary summary for couple of months for an employee, No-pay-value, base-pay-value, and gross pay value of all the employees for a given month range.

After entering the values and calculating. The calculated values will be displayed in the "Results of Calculations" section. Then the user can generate more values and print the desired report at the desired time. The data grid view can be used to edit the data as well.

## Coding For the Enter Button

```
private void button3_Click(object sender, EventArgs e)
     {
         con.Open();
         string query = "select * from settings";
         SqlCommand cmdn = new SqlCommand(query, con);
         DataTable dtr = new DataTable();
         SqlDataAdapter sdaa = new SqlDataAdapter(cmdn);
         sdaa.Fill(dtr);
         foreach (DataRow dr in dtr.Rows)
         {
             cyclbl.Text = dr["salcyc"].ToString();
             sdayp.Format = DateTimePickerFormat.Custom;
             sdayp.CustomFormat = "yyyy/MM/dd";
             sdayp.Text = dr["bdate"].ToString();
             edayp.Format = DateTimePickerFormat.Custom;
             edayp.CustomFormat = "yyyy/MM/dd";
             edayp.Text = dr["edate"].ToString();
             clbl.Text = dr["leaves"].ToString();
             govlbl.Text = dr["govrate"].ToString();
             cyclbl.Visible = true;
             sdayp.Visible = true;
             edayp.Visible = true;
             clbl.Visible = true;

         }

         string queryn = "select * from Employee where emp_id='" + empid.Text + "'";
         SqlCommand cmd = new SqlCommand(queryn, con);
         DataTable dt = new DataTable();
         SqlDataAdapter sda = new SqlDataAdapter(cmd);
         sda.Fill(dt);
         foreach (DataRow dr in dt.Rows)
         {


             mslbl.Text = dr["m_salary"].ToString();
             orlbl.Text = dr["orh"].ToString();
             allbl.Text = dr["allowances"].ToString();

             mslbl.Visible = true;
             orlbl.Visible = true;
             allbl.Visible = true;

         }
         con.Close();


     }
```

*Figure 16 - Coding For the Enter Button*

```
    private void Calculate()
{


    con.Open();

    decimal monthlySalary = decimal.Parse(mslbl.Text);
    decimal allowances = decimal.Parse(allbl.Text);
    decimal overtimeRate = decimal.Parse(orlbl.Text);
    decimal governmentTaxRate = decimal.Parse(govlbl.Text);
    int salaryCycleDays = int.Parse(cyclbl.Text);
    int noOfAbsentDays = int.Parse(abdays.Text);
    int noOfOvertimeHours = int.Parse(oh.Text);
    int noOfHolidays = int.Parse(holit.Text);

    // Calculate Base Pay
    decimal basePay = monthlySalary + allowances + (overtimeRate * noOfOvertimeHours);

    // Calculate No-pay Value
    decimal noPayValue = (monthlySalary / salaryCycleDays) * noOfAbsentDays;

    // Calculate Gross Pay
    decimal grossPay = (basePay - noPayValue) + (basePay * (governmentTaxRate/100));

    // Round off the calculated values to two decimal points
    basePay = Math.Round(basePay, 2);
    noPayValue = Math.Round(noPayValue, 2);
    grossPay = Math.Round(grossPay, 2);

    // Display the calculated values
    bplbl.Text = basePay.ToString();
    npvlbl.Text = noPayValue.ToString();
    gplbl.Text = grossPay.ToString();

    con.Close();

}
```

```
  private void button5_Click(object sender, EventArgs e)
        {
                Calculate();
                try
                {

                    sdayp.Format = DateTimePickerFormat.Custom;
                    sdayp.CustomFormat = "yyyy/MM/dd";
                    edayp.Format = DateTimePickerFormat.Custom;
                    edayp.CustomFormat = "yyyy/MM/dd";

                    con.Open();

                    string query = "insert into salaryt values('" + empid.Text + "','" + sdayp.Text +
"','" + edayp.Text + "','" + clbl.Text + "','" + abdays.Text + "','" + holit.Text + "','" + oh.Text +
"','" + npvlbl.Text + "','" + bplbl.Text + "','" + gplbl.Text + "')";
                    SqlCommand cmd = new SqlCommand(query, con);
                    cmd.ExecuteNonQuery();
                    MessageBox.Show("Your Data Has Beeen Successfully Added");
                    con.Close();
                    filldatagrid();

                }
                catch (Exception Ex)

                {

                    MessageBox.Show(Ex.Message);

                }
```

*Figure 17 – Calculate Button Code*

## Coding For The Print Button

```
SqlConnection con = new SqlConnection(@"Data Source=DESKTOP-EQH9955\SQLEXPRESS;Initial
Catalog=griffindo_toys;Integrated Security=True");
        private DataTable dataTable;

private void Form4_Load(object sender, EventArgs e)
        {
            filldatagrid();
            LoadData();
        }


    private void button1_Click(object sender, EventArgs e)
        {

            try
            {
                SaveFileDialog saveFileDialog = new SaveFileDialog();
                saveFileDialog.Filter = "CSV Files|*.csv";
                if (saveFileDialog.ShowDialog() == DialogResult.OK)
                {
                    string filePath = saveFileDialog.FileName;
                    ExportDataTableToCsv(dataTable, filePath);
                    MessageBox.Show("Data saved to " + filePath, "Success",
MessageBoxButtons.OK, MessageBoxIcon.Information);
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error: " + ex.Message);
            }

        }


        private void ExportDataTableToCsv(DataTable dataTable, string filePath)
        {
            using (System.IO.StreamWriter writer = new System.IO.StreamWriter(filePath))
            {

                string columnHeaders = string.Join(",",
dataTable.Columns.Cast<DataColumn>().Select(dc => dc.ColumnName));
                writer.WriteLine(columnHeaders);
                foreach (DataRow row in dataTable.Rows)
                {
                    string rowData = string.Join(",", row.ItemArray);
                    writer.WriteLine(rowData);
                }
            }
        }
```

```
private void LoadData()
{
    try
    {
        {
            con.Open();
            string query = "SELECT * FROM salaryt";
            SqlCommand cmd = new SqlCommand(query, con);

            SqlDataAdapter adapter = new SqlDataAdapter(cmd);
            dataTable = new DataTable();
            adapter.Fill(dataTable);
            dataGridView1.DataSource = dataTable;

            con.Close();
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}
```

*Figure 18 - The code to Print the reports in Salary Calculator*

## Data Grid view Code

```
1 reference
private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
    empid.Text = dataGridView1.SelectedRows[0].Cells[0].Value.ToString();
    sdayp.Text = dataGridView1.SelectedRows[0].Cells[1].Value.ToString();
    edayp.Text = dataGridView1.SelectedRows[0].Cells[2].Value.ToString();
    abdays.Text = dataGridView1.SelectedRows[0].Cells[6].Value.ToString();
    holit.Text = dataGridView1.SelectedRows[0].Cells[3].Value.ToString();
    oh.Text = dataGridView1.SelectedRows[0].Cells[5].Value.ToString();
    bplbl.Text = dataGridView1.SelectedRows[0].Cells[7].Value.ToString();
    gplbl.Text = dataGridView1.SelectedRows[0].Cells[8].Value.ToString();
    npvlbl.Text = dataGridView1.SelectedRows[0].Cells[9].Value.ToString();


}
```

*Figure 19 - Data Grid view Code*

## Clear Button Code (Delete all data in the temporary database table)

```
1 reference
private void button4_Click(object sender, EventArgs e)
{
    try
    {
        con.Open();
        string query = "DELETE FROM salaryt";
        SqlCommand cmd = new SqlCommand(query, con);
        int rowsAffected = cmd.ExecuteNonQuery();

        if (rowsAffected > 0)
        {
            MessageBox.Show(" Data Successfully Deleted");
        }
        else
        {
            MessageBox.Show("No records found in the table to delete.");
        }

        con.Close();
        filldatagrid();
    }
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred: " + ex.Message);
    }
}
```

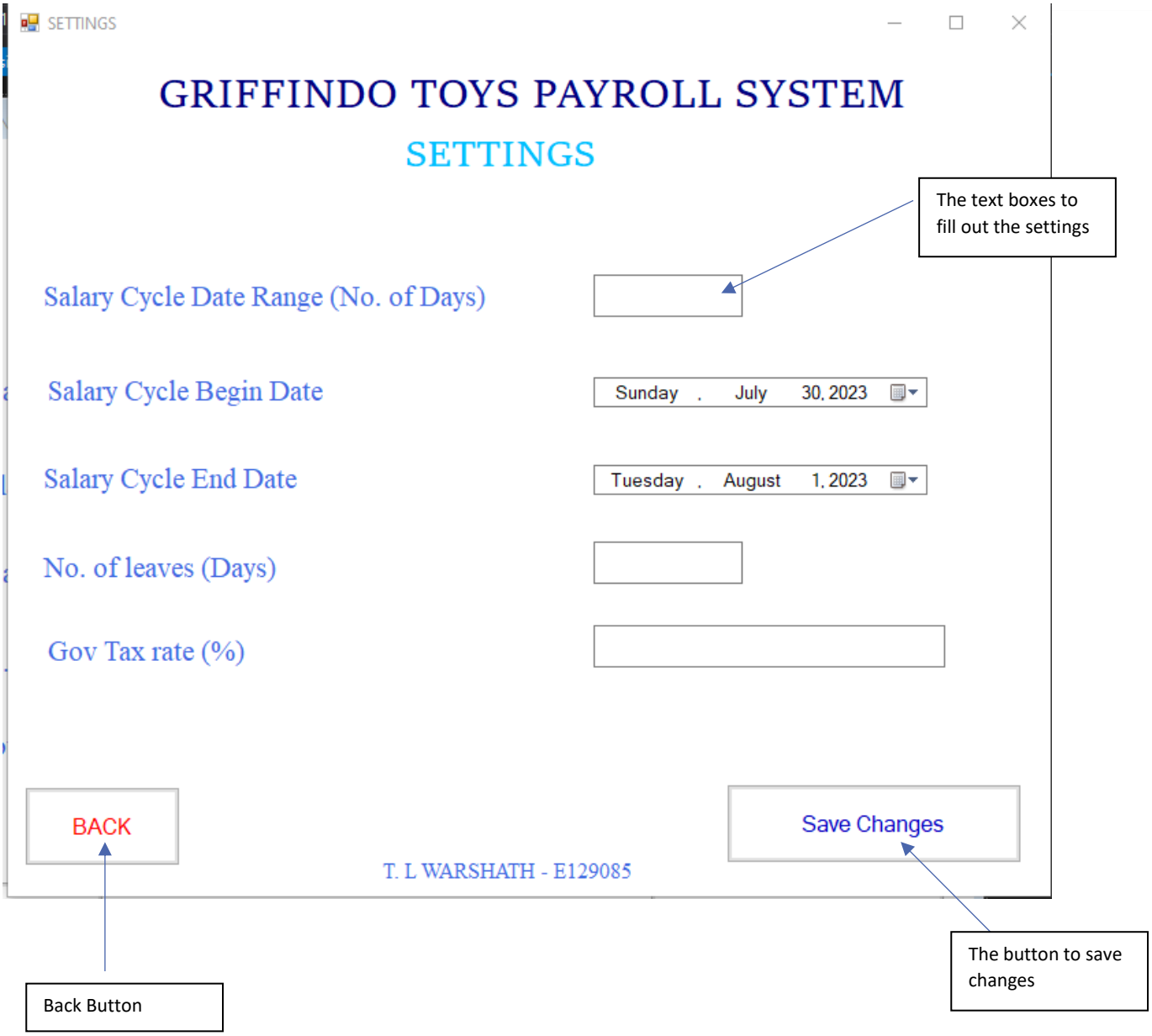
*Figure 20 - The clear Button Code*

## The Back Button Code

```
1 reference
private void button2_Click(object sender, EventArgs e)
{
    Form2 goo = new Form2();
    this.Hide();
    goo.Show();
}
```

*Figure 21 - The Back Button Code*

## The Settings Interface



The above interface is where the admin will be able to manage the default data which are required to the salary calculation of employees. The admin needs to simply enter the values and click “Save Changes” and the settings will be updated as desired.

**Code For the Save Changes Button**

```
        InitializeComponent();
    }
    SqlConnection con = new SqlConnection(@"Data Source=DESKTOP-EQH9955\SQLEXPRESS;Initial Catalog=griffindo_toys;Integrated Security=True");
    1 reference
    private void button2_Click(object sender, EventArgs e)
    {
        Form2 goo = new Form2();
        this.Hide();
        goo.Show();
    }

    1 reference
    private void button5_Click(object sender, EventArgs e)
    {
        try
        {
            sdate.Format = DateTimePickerFormat.Custom;
            sdate.CustomFormat = "yyyy/MM/dd";
            edate.Format = DateTimePickerFormat.Custom;
            edate.CustomFormat = "yyyy/MM/dd";

            con.Open();
            string query = "UPDATE settings SET salcyc=@salcyc, bdate=@bdate, edate=@edate, leaves=@leaves, govrate=@govrate WHERE setid=@setid";
            SqlCommand cmd = new SqlCommand(query, con);

            // Assuming 'leavet' is of type int, you can convert it to int before setting the parameter
            int leavesValue = Convert.ToInt32(leavet.Text);

            // Add parameters to the SqlCommand to prevent conversion issues
            cmd.Parameters.AddWithValue("@salcyc", salt.Text);
            cmd.Parameters.AddWithValue("@bdate", sdate.Text);
            cmd.Parameters.AddWithValue("@edate", edate.Text);
            cmd.Parameters.AddWithValue("@leaves", leavesValue); // Use the converted 'leavet' value here
            cmd.Parameters.AddWithValue("@govrate", govt.Text);
            cmd.Parameters.AddWithValue("@setid", 1); // Assuming 'setid' is of type int

            cmd.ExecuteNonQuery();
            MessageBox.Show("You Have Successfully Updated Settings");
            con.Close();
        }
        catch (Exception ex)
        {
            MessageBox.Show("An error occurred: " + ex.Message);
        }
    }
```

*Figure 22 – Save changes button code in settings*

**Back Button Code**

```
private void button2_Click(object sender, EventArgs e)
{
    Form2 goo = new Form2();
    this.Hide();
    goo.Show();
}
```

*Figure 23 - The code for the back Button*